

1. According to this tutorial (https://mmdetection3d.readthedocs.io/zh-cn/latest/get_started.html), build the conda environment of mmdetection3d.

set the file path in the create_data file.

```
parser.add_argument(
    '--root-path',
    type=str,
    default='/mnt/ssd1/yaojunguang/WorkSpace/3d_object_detection/data/nuscenes',
    # default='/mnt/ssd1/yaojunguang/WorkSpace/3d_object_detection/data/nus/v1.0-mini',
```

1. change the version to v1.0-mini

```
parser.add_argument(
    '--version',
    type=str,
    default='v1.0-mini',
    required=False,
    help='specify the dataset version, no need for kitti')
```

2. Then, in tools/nuscenes_converter.create_nuscenes_infos, to modify the scene information.

```
if version == 'v1.0-trainval':
    train_scenes = splits.train
    val_scenes = splits.train
    # val_scenes = ['scene-0061', 'scene-0103']
elif version == 'v1.0-test':
    train_scenes = splits.train
    val_scenes = []
elif version == 'v1.0-mini':
    train_scenes = splits.mini_train
    # val_scenes = splits.mini_val
    val_scenes = ['scene-0061', 'scene-0103']
    # val_scenes = splits.mini_train + splits.mini_val
    # print(train_scenes)
    # print(val_scenes)
```

3. Then, in tools/nuscenes_converter.fill_trainval_infos, set the lidar_path_list to the paths of the two lidar samples.

```
lidar_path_list = ['/mnt/ssd1/yaojunguang/WorkSpace/3d_object_detection/data/
nuscenes/samples/LIDAR_TOP/
n015-2018-07-24-11-22-45+0800__LIDAR_TOP__1532402937198682.pcd.bin',
'/mnt/ssd1/yaojunguang/WorkSpace/3d_object_detection/data/nuscenes/
samples/LIDAR_TOP/
n008-2018-08-01-15-16-36-0400__LIDAR_TOP__1533151613398020.pcd.bin']
```


Upsampling:

	original	RS_25	RS25_to100_GJ	RS25_to100_VI
mAP	0.2342	0.2122	0.2153	0.211
car	0.603	0.601	0.599	0.603
truck	0.43	0.315	0.309	0.315
bus	0	0	0	0
trailer	0	0	0	0
construction_vehicle	0.072	0.037	0.037	0.037
pedestrian	0.699	0.676	0.68	0.67
motorcycle	0.007	0	0.024	0
bicycle	0	0	0	0
traffic_cone	0.021	0.011	0.034	0.016
barrier	0.51	0.483	0.47	0.468
eval_time(s)	0.1s	0.1s	0.1s	0.1s

Upsampling:

	original	RS_50	RS50_to100_GJ	RS50_to100_VI
mAP	0.2342	0.2156	0.2232	0.2193
car	0.603	0.589	0.595	0.58
truck	0.43	0.361	0.4	0.361
bus	0	0	0	0
trailer	0	0	0	0
construction_vehicle	0.072	0.037	0.072	0.037
pedestrian	0.699	0.684	0.68	0.693
motorcycle	0.007	0.013	0.024	0.024
bicycle	0	0	0	0
traffic_cone	0.021	0.004	0.007	0.005
barrier	0.51	0.469	0.453	0.493
eval_time(s)	0.1s	0.1s	0.1s	0.1s